SECTION 2
NORMAL PROCEDURES

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LIST OF ILLUSTRATIONS

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Figure 2-1. Preflight check sequence 2–4

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SECTION 2
NORMAL PROCEDURES

INTRODUCTION

This section contains instructions and procedures for operating the helicopter from the planning stage, through actual flight conditions, to securing the helicopter after landing.

Normal and standard conditions are assumed in these procedures. Pertinent data in other sections is referenced when applicable.

The instructions and procedures contained herein are written for the purpose of standardization and are not applicable to all situations.

The minimum and maximum limits, and the normal and cautionary operating ranges for the helicopter and its subsystems are indicated by instrument markings and placards. Refer to Section 1 for a detailed explanation of each operating limitation.

Each time an operating limitation is exceeded, an appropriate entry shall be made in the logbook (helicopter, engine, etc.). The entry shall state which limit was exceeded, the duration of time, the extreme value attained, and any additional information essential in determining the maintenance action required.

FLIGHT PLANNING

Each flight should be planned adequately to ensure safe operations and to provide the pilot with the data to be used during flight.

Essential weight and balance, and performance information should be compiled as follows:

— Check type of mission to be performed and destination.
— Select appropriate performance charts to be used from Section 4.
— Review the appropriate Supplements of this Rotorcraft Flight Manual for the optional equipment(s) installed.

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TAKE-OFF AND LANDING DATA

Refer to Section 1 - LIMITATIONS and Section 4 - PERFORMANCE .

WEIGHT AND BALANCE DATA

Ascertain proper weight and balance of the helicopter as follows:

— Consult Section 6 - Weight and Balance.
— Ascertain weight of fuel, oil, payload, etc.
— Compute take-off and anticipated landing gross weights
— Check helicopter center of gravity (CG) locations.
— Check that the weight and CG limitations in Section 1 are not exceeded.

PREFLIGHT CHECKS

Preflight checks are intended as those checks to be performed by the pilot in order to ascertain that the helicopter is flightworthy and adequately equipped. They are therefore not meant as detailed mechanical inspections, but as a guide to check the condition of the helicopter.

″Check″ is intended as observing the helicopter to find any obvious damage. **″Damage″** is intended as any abnormal or out of limits condition.

If during preflight check these conditions are found, inspections shall be carried out before the flight, in order to ascertain the helicopter airworthiness.

Have care to brief passengers on relevant operational procedures and associated hazards.

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PILOT'S DAILY PREFLIGHT CHECK

(First flight of the day)

The following procedure outlines the pilot walk-around and interior checks (Figure 2-1).

Main and tail rotor tie-down (if present)	: Removed.

AREA N° 1 (Helicopter nose)

Nose exterior	: Condition.
Ventilation air intake	: Free of obstructions.
Landing lights	: Condition.
Nose landing gear	: Condition, shock strut extension, leaks, tire pressure.
Battery	: Connection secured.
Avionic equipment	: Secured.
Accumulators	: Condition and leaks. Discharge accumulators by pressing relevant red buttons. Door secured.

CAUTION

The discharge of accumulators causes loss of parking brakes.

Drains and breather	: Conditions.
Nose compartment access door	: Security.

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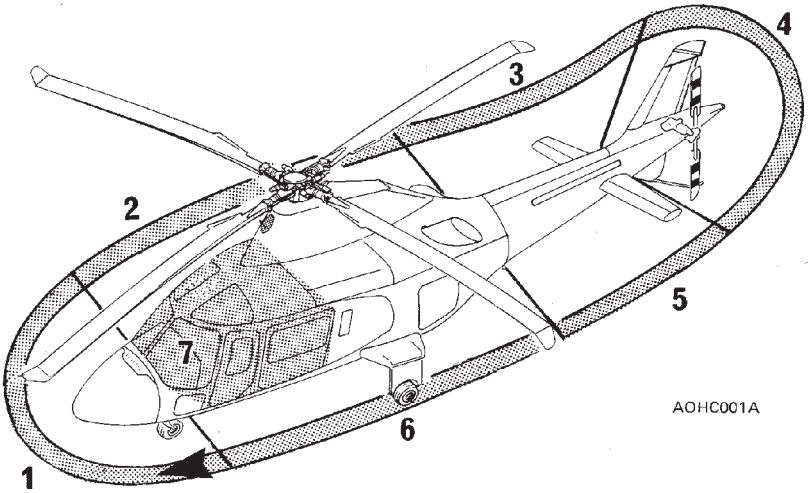

AREA No. 1 : Helicopter nose
AREA No. 2 : Fuselage - RH side
AREA No. 3 : Tail boom - RH side
AREA No. 4 : Fins, 90° gearbox, tail rotor, tail skid
AREA No. 5 : Tail boom - LH side
AREA No. 6 : Fuselage - LH side
AREA No. 7 : Cabin interior

Figure 2-1. Preflight check sequence.

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AREA N° 2 (Fuselage - rh side)

Windshield and roof transparent panel	: Condition and cleanliness.
Pitot tube	: Cover removed, condition and obstructions.
IDS external air temperature probe	: Condition, free of obstructions.
Fuselage exterior	: Condition.
Pilot cabin door	: Condition, cleanliness, window secured.
Antenna(s)	: Condition.
Passenger cabin door	: Condition, cleanliness, window secured.
Passenger door jettison window	: Security of window and seal retainer.
Cowling and fairings	: Condition and security.
Servo hydraulic system valves and filter group	: Check for leaks and by-pass indication (red button out: filter clogged). Door secured.
Hydraulic system tanks	: Check fluid level and filler cap for security. Door secured.
Main rotor hub and blades	: Condition and security.
Main rotor dampers	: Check for correct charge indication.

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Main rotor pitch change links	: Condition and security.
Upper anticollision light	: Condition.
Servo actuator	: Condition and leaks.
Main transmission and accessories (visible area)	: Condition and leaks.
Transmission oil cooler	: Free of obstructions.
Transmission external oil filter	: By-pass indication (white button out: filter clogged). Door secured.
Cooler blower air intake	: Free of obstruction.
Cooler system belt	: Condition and security.
Airframe (A/F) fuel filter	: Condition and leaks.
Service step	: Secured.
Engine air intake screen and chamber	: Cover removed; free of damage and/or obstructions.
Engine area	: Check for fuel and/or oil leaks.
Engine oil	: Check gauge for oil presence (sight glass dark).
Engine oil filter impending bypass indicator	: Check for correct indication.
Engine-transmission drive shaft	: Condition.
Engine supports (visible area)	: Condition.

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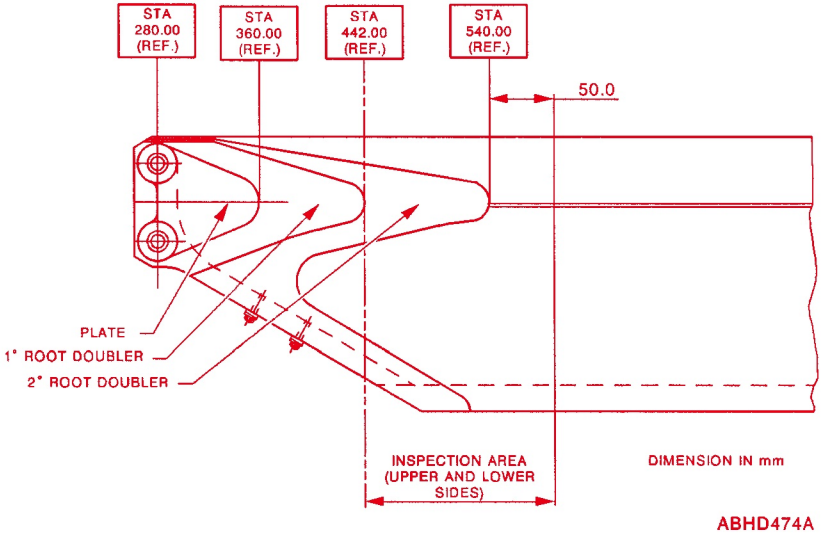
Add the following step in paragraph "AREA N°4 (Fins, tail gearbox, tail rotor and skid)":

Tail rotor blades
(P/N 109-8132-01-111) : Check for absence of cracks on upper and lower sides. (Refer to figure herewith enclosed).

NOTE

Paint irregularities may be due to cracks.



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Engine cowling	: Condition, secured.
Fuel filler cap	: Secured.
Main landing gear	: Condition, shock strut extension, leaks, tire pressure.
Drain and vent lines	: Free of obstructions.
Lower anticollision light	: Condition.
Antenna(s) (if installed)	: Condition.
Engine exhaust	: Cover removed, condition.

AREA N° 3 (Tail boom - rh side)

Tail boom exterior	: Condition.
Antenna(s) (if installed)	: Condition.
Stabilizer	: Condition and security.
Navigation light	: Condition.

AREA N° 4 (Fins, tail gearbox, tail rotor and skid)

Upper tail fin	: Condition.
Lower tail fin and skid	: Condition.
Tail rotor long drive shaft bearing	: Check condition verifying no mark slippage. Door secured.
Tail rotor gearbox	: Check oil level, check for leaks and for filler cap secured.

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Access door	:	Secured.
Tail navigation light	:	Condition.
Tail rotor hub and blades	:	Condition, cleanliness and freedom of flapping.
Tail rotor pitch change mechanism	:	Condition and security.

Area n° 5 (tail boom - lh side)

Tail boom exterior	:	Condition.
Stabilizer	:	Condition and security.
Navigation light	:	Condition.
Antenna(s) (if installed)	:	Condition.
Tail rotor long drive shaft bearings	:	Check condition verifying no mark slippage.
Tail rotor drive shaft cover	:	Secured.

AREA N° 6 (Fuselage - lh side)

Fuselage exterior	:	Condition.
Engine exhaust	:	Cover removed, condition.
Baggage compartment	:	Cargo (if on board) properly secured.
Power relay box circuit breakers	:	In. Access door secured.

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NOTE

The above mentioned circuit breakers are accessible from baggage compartment through an inspection door.

Baggage door	: Secured.
Lower avionic bay access door	: Secured.
Tail rotor middle drive shaft bearings	: Check condition verifying no mark slippage. Door secured.
Engine area	: Check for fuel and/or oil leaks.
Engine oil	: Check gauge for oil presence (sight glass dark).
Engine oil filter impending bypass indicator	: Check for correct indication.
Engine-transmission drive shaft	: Condition.
Engine supports (visible area)	: Condition.
Engine cowling	: Condition, secured.
Engine air intake screen and chamber	: Cover removed; free of damage and/or obstructions.
Cooler system belt	: Condition and security.
Main rotor hub and blades	: Condition and security.
Main rotor dampers	: Check for correct charge indication.
Main rotor pitch change links	: Condition and security.

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Upper anticollision light	: Condition.
Swashplate and driving scissors	: Condition and security.
Main transmission and accessories (visible area)	: Condition and leaks.
Transmission	: Filler cap secured.
Transmission oil	: Check correct level. Door secured.

NOTE

If rotor brake (if installed) has been used, the oil level indication could be lower than actual level. Therefore, when oil level is below minimum level mark, before replenishing transmission, it is necessary to shut down engines without operating rotor brake, in order to determine correct amount of oil required to top up the transmission.

Servo actuators	: Condition and leaks. Door secured.
Cooler blower air intake	: Free of obstructions.
Airframe (A/F) fuel filter	: Condition and leaks.
Service step	: Secured.
Drain and vent lines	: Free of obstructions.
Lower avionic bay access door	: Secured.
Main landing gear	: Condition, shock strut extension, leaks, tire pressure.
Passenger door jettison window	: Security of window and seal retainer.

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Passenger cabin door	:	Condition, cleanliness, window secured.
Cowling and fairings	:	Condition and security.
Copilot cabin door	:	Condition, cleanliness, window secured.
Antenna(s)	:	Condition.
Pitot tube	:	Cover removed, condition and obstructions.
Windshield and roof transparent panel	:	Condition and cleanliness.

AREA N°7 (Cabin interior)

Passenger jettison window	:	Security.
Cabin interior	:	Security of equipment and cargo.
First aid kit	:	On board.
Cabin fire extinguisher	:	Security.
Passenger doors	:	Security. Sliding windows closed, if installed.
Copilot door jettison handle and relevant safety latch	:	Correct position.
Copilot safety belt and inertia reel	:	Condition and belt fastened if seat is unoccupied.
Copilot seat	:	Security.

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Copilot flight controls (if installed)	: Condition and security.
Landing gear lever lock	: Check in NORMAL position and lockwired.
Lower and lateral transparent panels	: Cleanliness and integrity.
Copilot door	: Security. Sliding window closed, if installed.
BATT RELAY circuit breaker	: In.

NOTE

The above mentioned circuit breaker is accessible from the pilot pedal bay and thus cannot be seen from the on-board seated position.

CVDR TEST (if installed)

BATT	: ON.
FDR and CVR fault lights	: After 5 seconds, check both lights go off.
CVDR test	: Press and hold TEST switch on the CVDR control panel for at least 5 seconds. Check green lamp TEST stays on (the TEST lamp stays on as long as the TEST button is pressed).
BATT	: OFF.
Pilot door	: Security. Sliding window closed, if installed
Pilot door jettison handle and relevant safety latch	: Correct position.
Pilot safety belt and inertia reel	: Condition.

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Pilot seat	:	Security.
Pilot flight controls	:	Condition and security.
Lower and lateral transparent panels	:	Cleanliness and integrity.
Instruments, panels and breakers	:	Condition and legibility.
ELT switch on instrument panel (if installed)	:	Check in ARM position.

For the following checks connect the d.c. supply.

Accumulators (HYDRAULIC UTILITY NORM/EMER)	:	Check for correct precharge.
LH airframe (A/F) fuel filter	:	Gently drain while respective fuel pump is operating. Check for by-pass indication, # 1 A/F F FLTR caution message on EDU 1, by pushing red button on filter.

NOTE

Fuel is pressurized, therefore drainage should be carried out by gently pushing red button. Failure to comply with this advice could lead to some fuel being squirted around.

LH fuel drain switch	:	Raise the guard and operate the switch to drain the respective fuel sump. Check for fuel dripping from the drain and FUEL DRAIN 1 caution message displayed. Fully lower the guard to switch the drain valve off. Check FUEL DRAIN 1 caution message out and no more fuel dripping.

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RH airframe (A/F) fuel filter	: Gently drain while respective fuel pump is operating. Check for by-pass indication, # 2 A/F F FLTR caution message on EDU 1, by pushing red button on filter.

NOTE

Fuel is pressurized, therefore drainage should be carried out by gently pushing red button. Failure to comply with this advice could lead to some fuel being squirted around.

RH fuel drain switch	: Raise the guard and operate the switch to drain the respective fuel sump. Check for fuel dripping from the drain and FUEL DRAIN 2 caution message displayed. Fully lower the guard to switch the drain valve off. Check FUEL DRAIN 2 caution message out and no more fuel dripping.

Check following systems for correct operation:
— Navigation and anticollision lights.
— Landing light.

Disconnect the d.c. electrical supply.

PILOT'S PREFLIGHT CHECK

(Every flight)

Main and tail rotors tie-down (if present)	: Removed.
Nose, fuselage, tail boom exterior, tail fins and skid	: Check.

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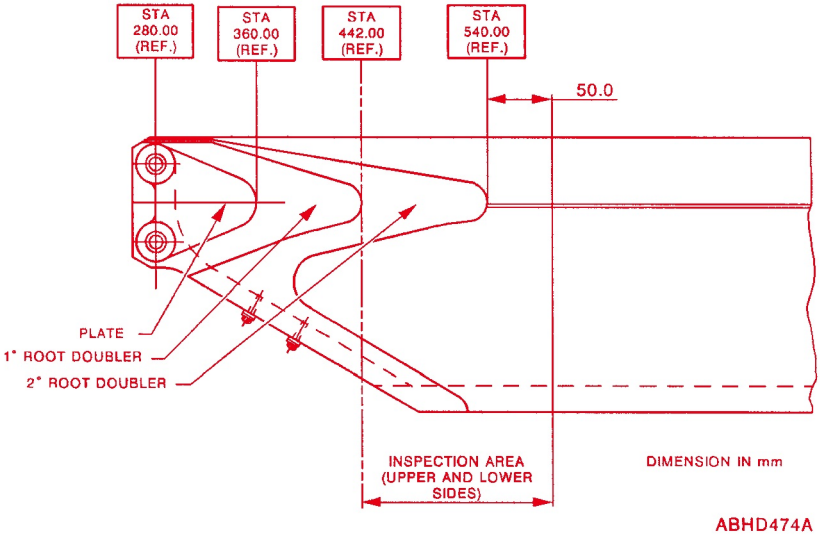
Add the following step in paragraph "PILOT'S PREFLIGHT CHECK (Every flight)":

Tail rotor blades
(P/N 109-8132-01-111) : Check for absence of cracks on upper and lower sides. (Refer to figure herewith enclosed).

NOTE

Paint irregularities may be due to cracks.



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Stabilizers	:	Condition and security.
Nose and main landing gear	:	Check.
Access doors nose compartment, tail rotor gearbox, baggage compartment, lower avionic bay	:	Security.
Cowlings, fairings and relevant access doors	:	Condition and security, check for evidence of oil.
Windshield and roof transparent panels	:	Condition and cleanliness.
Pitot tubes	:	Free of obstructions. Covers removed.
Crew and passenger cabin doors	:	Condition, cleanliness, windows secured, doors secured, latches in correct position. Sliding windows closed, if installed.

CVDR TEST (if installed)

BATT	:	ON.
FDR and CVR fault lights	:	After 5 seconds, check both lights go off.
CVDR test	:	Press and hold TEST switch on the CVDR control panel for at least 5 seconds. Check green lamp TEST stays on (the TEST lamp stays on as long as the TEST button is pressed).
BATT	:	OFF.

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Cabin interior and baggage compartment	:	Security of equipment and cargo.
Main and tail rotor blades	:	Check.

FOR TRAINING ONLY

Air intakes	: Free of obstructions. Covers removed.
Engine exhausts	: Check. Covers removed.
Fuel filler cap	: Secured.
Crew safety belts and inertia reels	: Condition, belt fastened.
Tail rotor gearbox	: Check oil level, check for leaks and for filler cap secured.
Transmision oil	: Check correct oil level. Door secured.

NOTE

If rotor brake (if installed) has been used, the oil level indication could be lower than actual level. Therefore, when oil level is below minimum level mark, before replenishing transmission, it is necessary to shut down engines without operating rotor brake, in order to determine correct amount of oil required to top up the transmission.

ENGINE PRE-START CHECK

All switches	: OFF or CLOSED.
Pedals and seat	: Adjust.
Seat belt	: Fasten and adjust.
Nose wheel lock	: ON (lever up).
Parking brake	: ON (pull out and turn).
Cyclic stick	: Centered (or positioned to counter wind) and friction adjusted.

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Collective lever	: Full down and friction adjusted.
Circuit breakers	: All engaged.
Engine power levers	: OFF (full aft).
ENG 1 and 2 MODE switches	: OFF (fully counterclockwise).
LD-SH (Load share) switch	: As required.

NOTE

The LD - SH switch allows the pilot to keep matched engine torque or TOT, as required.

OEI MODE switch (if installed)	: NORM, check.
OEI TNG switch (if installed)	: Off position (centered), check.
Altimeter	: Set.
STATIC source switch	: NORM and protected.
LDG GEAR control lever	: DOWN, check.
SERVO (Main hydraulic)	: NORM.
SAS 1 and 2 switches	: OFF.
AUTOTRIM switch	: OFF.

NOTE

The AUTOTRIM switch should be turned off during ground operation to preclude undesired cyclic movements.

External power (if used)	: Connect.

NOTE

Be sure that the external power source supplies not less than 28 volt.

FOR TRAINING ONLY

NOTE

(Only for helicopters equipped with emergency bus).
The battery is automatically disconnect when an external power source is connected to the helicopter. Check that BATT OFF caution message is displayed on EDU 1 if the external power is used.

BATTery : ON.

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GEN BUS 1 and 2	: ON.
GEN 1 and 2	: ON
Force trim	: ON.
Electronic Display Units (EDU 1 and EDU 2)	: Check on.

NOTE

Both EDUs are automatically activated when the helicopter is electrically powered: EDU 1 in CRUISE mode and EDU 2 in MAIN mode.

EDU 1 : Select MENU, press TEST key and check the following test sequences on both EDUs.

NOTE

The TEST function can also be selected on EDU 2.

EDU 1 : The test sequence shall display the CRUISE format with the following data:

N1	97.4%
TOT	820°C
TRQ	100%
NR/N2	101%

The caution and warning messages, stimulated by the DAU, shall be presented in the message area during test routine.

NOTE

During the test the DAU will activate the MASTER WARNING/MASTER CAUTION lights, the ENG FIRE warning message and it will also illuminate the engine power lever grips. If a failure is detected on engine fire and/or fuel low detectors, the caution message FIRE DET and/or F LOW FAIL will appear.

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NOTE

It is possible to listen to only one of the ENG FIRE messages due to the short length of the EDU TEST.
The ENG FIRE message of the second engine can be heard, even during the test sequence, by pushing the MASTER WARNING RESET pushbutton.

NOTE

If the helicopter is configured with the engine control panel P/N 109-0900-55-101 on the central console, the test of the EDU will also activate the FIRE warning lights adjacent to the engine mode switches.

EDU 2 : The test sequence shall display the MAIN format with the following data:

ENG OIL pressure	50 PSI
ENG OIL temperature	100°C
XMSN OIL pressure	40 PSI
XMSN OIL temperature	100°C
FUEL pressure	20 PSI
Hydraulic oil pressure	1500 PSI

Fuel quantity, decreasing.

In the advisory area (lower part of the screen) the test sequence shall display:
EDU 1 and EDU 2 software identification number.
DAU-A and -B software identification number.
EDU 1 BIT PASS.
EDU 2 BIT PASS.
DAU-A BIT PASS.
DAU-B BIT PASS.

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NOTE

In case of different result of TEST sequence refer to pertinent paragraph in Section 3.

EDU 1 and EDU 2	:	After 10 seconds automatically return to previous selected formats.
EDU 1	:	Select MENU and enter page 2. Verify CH-A and CH-B legends to be green and white respectively.
Aural Warning Generator Test	:	Set AWG switch on TEST and maintain. Check the aural message ″TEST OK″ and after about 6 seconds the vocal alarm operates in the following sequence: ROTOR LOW ENGINE 1 OUT ENGINE 2 OUT ENGINE 1 FIRE ENGINE 2 FIRE WARNING ROTOR HIGH LANDING GEAR 150 FEET

NOTE

The vocal alarm message ″WARNING″ is activated when # 1 ECU FAIL, # 2 ECU FAIL, XMSN OIL PRES, XMSN OIL HOT or BATT HOT are displayed on EDU 1.

NOTE

The test sequence also includes the ″200 FEET″ message. This message is not utilized by the system and it is only produced during the test, being a foreseen option of the Aural Warning Generator.

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ADI stand-by	: Flag retracted, erect (operate ″PULL TO CAGE″ knob, if necessary).
Fuel quantity	: Check.
RPM switch (on collective control)	: Set 100%.
Engine power levers	: Set gently to FLIGHT position.

CAUTION

Avoid rapid movement of engine power levers in order to not damage the internal mechanism.

# 1 ENG GOV switch	: AUTO.
# 2 ENG GOV switch	: AUTO.
ENG TRIM toggle switches (on collective control)	: Verify the operation, then leave the engine power levers in the FLIGHT position

NOTE

Each engine trim toggle switch controls the respective power lever from IDLE to FLIGHT position when in AUTO mode, and from IDLE to MAX position when in MANUAL mode.

NOTE

Both engine power levers should always be operated through the toggle switches located on the collective control stick. They shall be operated manually only in case of failure of the remote control (PLA MOTOR caution message active), or before starting, to position the levers to FLIGHT.

ENG 1 FUEL switch	: ON - Fuel valve indicator, vertical.

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FUEL PUMP 1 switch	: ON - FUEL PUMP 1 caution message out, check pressure.
ENG 2 FUEL switch	: ON - Fuel valve indicator vertical.
FUEL PUMP 2 switch	: ON - FUEL PUMP 2 caution message out, check pressure.

NORMAL ENGINE START

Either engine may be started first.
Either engine may be started using either the auto or the manual mode.

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NOTE

It is recommended the normal engine starts be made using the auto mode. For starting procedure in manual mode refer to Section 3.

Collective control	: Flat pitch, check.
Rotor brake (if installed)	: Disengaged (lever full forward).
EDU 1	: Select START page.

ENGINE 1 START

ENG 1 MODE switch	: IDLE.

NOTE

(only for helicopters equipped with emergency bus)
If the battery discharge control box is installed after few seconds the BATT DISCH warning message on EDU1 will be displayed.

NOTE

It is recommended to start the engine to IDLE, nevertheless, if necessary, it is possible to start to FLIGHT by setting the ENG MODE switch directly to FLT.

Gas Producer (N1)	: Note increasing and START legend vertically displayed.
Engine temperature (TOT)	: Note increasing and IGN legend vertically displayed.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limit
- N1 or N2 increases beyond engine limits
- engine hangs (stagnation in N1 below 54%)

shutdown the engine by setting OFF the pertinent ENG MODE switch.

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CAUTION

Following aborted start shutdown perform the following procedure before restarting:
- Allow 30 seconds fuel drain period
- Perform a 30 seconds DRY MOTORING RUN.

Refer to Section 1 for engine starter limitations and to DRY MOTORING RUN procedure in this Section.

Engine oil pressure : Check.

NOTE

During cold starting conditions the engine oil pressure can rise up to a transient of 200 PSI. The pressure decreases as oil temperature rises. The oil pressure limits in Section 1, Fig. 1-5 must be respected when main oil temperature is 71°C - 120°C.

CAUTION

Do not apply power until engine oil temperature reaches 10°C.

WARNING

If the main rotor has not begun to rotate when gas producer (N1) reaches 40%, abort the start by setting the ENG MODE switch to OFF.

Engine N°1 starter : Automatically deactivated when N1 is 50%.
START and IGN legends automatically suppressed.

Main hydraulic system : When the main rotor begins to rotate, check rise in main hydraulic pressure.

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Hydraulic utility system (normal and emergency) : When accumulators are discharged, as main rotor begins to rotate, check pressure rise in both systems and note the activation of MAIN UTIL CHRG and EMER UTIL CHRG caution messages. Note both caution messages are suppressed when systems are charged.

CAUTION

If MAIN UTIL CHRG and/or EMER UTIL CHRG caution messages are not displayed during systems charging, abort and correct trouble.

N°1 engine power turbine speed (N2) : Check stabilized to IDLE speed of 65% ± 1%.

NOTE

If the engine has been started directly to FLT (flight) the N2 will stabilize to 100%.

NOTE

Avoid any cyclic movement except to prevent hitting blade stops below 85% rotor RPM.

Engine and transmission oil : Check pressure and temperature.

ENG 1 MODE switch : FLT

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NOTE

In the starting phase it is suggested to select FLIGHT mode as soon as possible in order to speed up the engine oil heating.

ENGINE 2 START

Repeat above procedure to start engine N°2.

CAUTION

Ensure that the second engine engages as the N2 reaches FLIGHT condition. A nonengaged engine shows positive N2 value and near zero torque. If a nonengagement occurs, shut down the nonengaged engine first. When the nonengaged engine has stopped, shut down the engaged engine. If a sudden (hard) engagement occurs, shut down both engines. Maintenance action is required.

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Engine parameters	: Check within limits.
External power	: Disconnect (if used). Check BATT OFF caution message out (only for helicopters equipped with emergency bus).
INV 1 and 2 switches	: ON.
RAD- MSTR switch	: ON.
Clock	: Set.
Rotor speed (NR)	: 100% check.

QUICK ENGINE START

This procedure can be followed whenever the situation requires to speed up the takeoff.

CAUTION

It is not advised to accomplish this procedure on battery.

Collective control	: Flat pitch, check.
Rotor brake (if installed)	: Disengaged (lever full forward).
EDU 1	: Select START page.
ENG 1 MODE switch	: FLT.
ENG 2 MODE switch	: FLT, when engine 1 gas producer (N1) is above 10%.

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CAUTION

Avoid to operate the ENG MODE switches simultaneously.

Gas producer (N1)	: Note increasing and START legend vertically displayed.
Engine Temperature (TOT)	: Note increasing and IGN legend vertically displayed.

CAUTION

Monitor engine start and if any of the following occurs:
- lightup is not obtained within 10 seconds of fuel on
- abnormal noises are heard
- TOT increases beyond start limit
- N1 or N2 increases beyond engine limits
- engine hangs (stagnation in N1 below 54%)

shutdown the engine by setting OFF the pertinent ENG MODE switch.

CAUTION

Following aborted start shutdown perform the following procedure before restarting:
- Allow 30 seconds fuel drain period
- Perform a 30 seconds DRY MOTORING RUN.

Refer to Section 1 for engine starter limitations and to DRY MOTORING RUN procedure in this Section.

Engine oil pressure	: Check.

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NOTE

During cold starting conditions the engine oil pressure can rise up to a transient of 200 PSI. The pressure decreases as oil temperature rises. The oil pressure limits in Section 1, Fig. 1-5 must be respected when main oil temperature is 71°C - 120°C.

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CAUTION

Do not apply power until engine oil temperature reaches 10°C.

WARNING

If the main rotor has not begun to rotate when gas producer (N1) reaches 40%, abort the start by setting the ENG MODE switch to OFF.

Engine N°1 and N°2 starters	: Automatically deactivated when N1 are 50%. START and IGN legends automatically suppressed.
Main hydraulic system	: When the main rotor begins to rotate, check rise in main hydraulic pressure.
Hydraulic utility system (normal and emergency)	: When accumulators are discharged, as main rotor begins to rotate, check pressure rise in both systems and note the activation of MAIN UTIL CHRG and EMER UTIL CHRG caution messages. Note both caution messages are suppressed when systems are charged.

CAUTION

If MAIN UTIL CHRG and/or EMER UTIL CHRG caution messages are not displayed during systems charging, abort and correct trouble.

FOR TRAINING ONLY

Engine power turbine speed (N2)	:	Check both stabilized to Flight speed, 100%.

NOTE

Avoid any cyclic movement except to prevent hitting blade stops below 85% rotor RPM.

Engine and transmission parameters	:	Check within limits.
External power	:	Disconnect (if used). Check BATT OFF caution message out (only for helicopters equipped with emergency bus).
INV 1 and 2 switches	:	ON.
RAD-MSTR switch	:	ON.
Clock	:	Set.
Rotor speed (NR)	:	100% check.

FALSE START

DRY MOTORING RUN

NOTE

The following procedure is used to clear internally trapped fuel and vapor or if there is evidence of a fire within the engine.

ENG GOV switch	:	AUTO or MANUAL.
ENG MODE switch	:	OFF.
Power control lever	:	OFF.
ENG FUEL switch	:	OFF.
Fuel pump	:	OFF.
STARTER GEN IGN circuit breaker	:	Out.

FOR TRAINING ONLY

NOTE

When STARTER GEN IGN circuit breaker is out, the pertinent ignition system is deactivated even though the Engine Control Unit will provide the IGN message to the EDU.

Starter pushbutton (on engine power lever) : Push and hold as necessary.

NOTE

To operate the starter it is also possible to utilize the ENG MODE switch by setting IDLE.

CAUTION

Refer to Section 1 for engine starting limitations.

Gas producer (N1) : Note increasing.

Starter pushbutton : Release.

STARTER GEN IGN circuit breaker : In.

SYSTEMS CHECK

Engine and transmission oil : Pressure and temperature within limits.

NOTE

The transmission oil pressure can be in the cautionary range (yellow band) provided that the oil temperature is below 65°C.

SERVO (Main Hydraulic) : NORM, check. Make small clockwise cyclic movements, collective and pedals movements. Pressure drops must be equal for both systems (N°1 and N°2) and they should not exceed 70 psi.

FOR TRAINING ONLY

Set 2 OFF. SERVO 2 caution message displayed on EDU 1.
Check operation of system N°1 with same cyclic, collective and pedal movements.
Pressure drop should not exceed 70 psi and there should be no force increase, discontinuity or cyclic/collective coupling.

Repeat check with system N°1 OFF to check system N°2.

Then set NORM.

NOTE

Tail rotor boost pressure is furnished by system N°1. When system N°2 is being checked, it is normal that the rotor control pedals will be unboosted.

CROSS FEED switch	:	NORM (bar vertical)
FUEL PUMP 1	:	OFF. Note fall in fuel pressure, activation of FUEL PUMP 1 caution message on EDU 1, automatic operation of crossfeed valve (bar horizontal), XFEED advisory message on EDU 2 activated and consequent new increase of fuel pressure.

NOTE

When FUEL PUMP 1 (2) is OFF and crossfeed valve is active, N°2 (1) FUEL quantity box shall appear in magenta.

FOR TRAINING ONLY

FUEL PUMP 1	:	ON. FUEL PUMP 1 caution message out and crossfeed valve automatically closed (bar vertical) and XFEED advisory message suppressed.
FUEL PUMP 2	:	OFF. Note fall in fuel pressure, activation of FUEL PUMP 2 caution message on EDU 1, automatic operation of cross-feed valve (bar horizontal), XFEED advisory message on EDU 2 activated and consequent new increase of fuel pressure.
FUEL PUMP 1	:	OFF. Note fall in fuel pressure, activation of FUEL PUMP 1 and 2 caution messages, cross-feed valve still open (bar horizontal) and XFEED message still present. Verify operation of engine driven fuel pumps.
FUEL PUMP 1 and 2	:	ON. FUEL PUMP 1 and 2 caution messages out and crossfeed valve automatically closed (bar vertical) and XFEED message suppressed.
PITOT heat	:	ON (singly), check current peak on loadmeter and PITOT 1 (2) HEAT advisory message displayed; then OFF.
INVerter 1	:	OFF. Check INV 1 caution message displayed. Check for proper reading (115 V) on both AC systems on EDU 2 AUX display.

FOR TRAINING ONLY

INVerter 1	: ON. Check INV 1 caution goes out.
INVerter 2	: OFF. Check INV 2 caution message displayed. Check for proper reading (115 V) on both AC systems on EDU 2 AUX display.
INVerter 2	: ON. Check INV 2 caution goes out.
Cyclic stick	: Friction fully unlocked, freedom of movement.
COMPASS switch	: MAG.

Set communication and navigation frequencies as required and check audio panel.

Marker beacon lights	: Test.
IVSI	: Needle near zero. Vertical speed selector as desired.
Altimeters	: Set and check.

Gyro Compass (HSI instrument):

Compass flag	: Retracted.
Compass heading	: Consistent with MAG COMP HDG.
Select heading	: As desired.
Select nav. course	: As desired.

ADI stand-by	: Erected and flag retracted. Verify the consistency with the ADI.
ADI	: Erected and flag retracted. Verify the consistency with the ADI stand-by.
Hydraulic system	: Check.

FOR TRAINING ONLY

Flight director (if installed):

SBY button	: Press and check - SBY pushbutton: illuminated - ALL mode controller lights: on - GA and DH lights on ADI: on. - FD flag on ADI: in view.
GA button	: Press and check - Pushbutton (ON): illuminated. - GA light on ADI: on. - Pitch Cue: visible and centered. - Roll Cue: visible and centered. - Collective Cue: visible and below center.
HDG button	: Press and check - Pushbuton (ON): illuminated. - Roll Cue should follow movement of heading, select ″bug″ (HDG on HSI instrument).
FD/SBY switch (on cyclic stick)	: Press - FD should return to stand-by.

Radio Altimeter (if installed):

Indication	: Zero altitude.
OFF flag	: Retracted.
DH SET selector	: Set 50 ft.
TEST switch	: Press and maintain - DH light: out. - Needle 100 ft.
TEST switch	: Release - DH light: on.

FOR TRAINING ONLY

HELIPILOT system:

SAS 2 switch	: SAS 2: - SAS 2 OFF caution message out. - ATTD OFF caution message out.

Cyclic stick motion must not cause HELIPILOT indicator motion.

SAS 2 switch	: OFF: - SAS 2 OFF caution message displayed.
SAS 1 switch	: SAS 1: - SAS 1 OFF caution message out. - ATTD OFF caution message out.

Cyclic stick motion must not cause HELIPILOT indicator motion.
Pedal motion will cause YAW HELIPILOT indicator motion.

SAS 1 switch	: OFF: - SAS 1 OFF caution message displayed.
SAS 1 switch	: SAS 1: - SAS 1 OFF caution message out.
SAS 2 switch	: SAS 2: - SAS 2 OFF caution message out.
ATTD HOLD switch	: ATTD HOLD: - ATTD OFF caution message out.
FD controller (if installed)	: Stand-by (SBY).

FOR TRAINING ONLY

COUPLED/DECOUPLED switch (if installed)	: As desired.
Autotrim (if installed):	
AUTOTRIM switch	: AUTOTRIM.
HELIPILOT indicators	: Check centered.
Ammeters	: Within limits and sharing load.
Voltmeters	: Within limits.
Cockpit lights	: As required.
External lights	: Check and leave as required.
PITOT 1 and 2 switches	: ON.
PITOT 1(2) FAIL caution light (if installed)	: Push to test.
PITOT 1 and 2 switches	: Leave as required.

CAUTION

Turn Pitot heat on for flight in visible moisture and rain regardless of ambient temperature.

Power levers	: Check FLIGHT position (no #1 (#2) PLA message displayed).
RPM switch	: Set 102%.
NR/N2 speed	: 102% stabilized, check.
Parking brake	: Off. Check PARK BRK ON caution message out.
Caution and warning messages	: Check none.

FOR TRAINING ONLY

TAXIING

Nose wheel lock	:	OFF.
Collective and cyclic	:	Increase slowly the collective then move the cyclic stick forward moderately to start movement.
Pedal brakes	:	Check operation.
Pedal control	:	As required to select the direction.
Collective and pedal brakes	:	To reduce speed and to stop, lower the collective and apply pedal brakes.
Nose wheel lock	:	ON.

NOTE

If the nose wheel is not aligned forward it will be self-centered and locked as soon as the helicopter lifts off.

TAKE-OFF

HOVER TAKE-OFF

Collective pitch	:	Increase slowly and lift the helicopter to a hover.

CAUTION

The helicopter is free of ground resonance.
However if, for some reason, ground resonance should occur, lift the helicopter free of the ground immediately. If unable to become airborne, lower collective and shut-down engines.

FOR TRAINING ONLY

Tail rotor pitch	:	Apply as necessary to maintain direction.
Flight instruments	:	Check.
Engine instruments	:	Within limits.
Main hydraulic system instruments	:	Within limits.
Utility hydraulic system instruments	:	Within limits.

NOTE

In hover the helicopter has a slight inclination to the left. During lift-off correct as necessary.

Cyclic and collective	:	Apply as necessary to accelerate and climb smoothly, reaching take-off power at approximately 60 knots.
Landing gear	:	UP (by 200 ft AGL). Check light sequence: - Green light off - Red light on (retraction) - Red light off (retracted and locked)

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

MAIN UTIL CHRG caution message	:	Displayed during landing gear operation. Out after the landing gear is locked.
Utility hydraulic systems	:	Pressure within limits.

FOR TRAINING ONLY

FTR pushbutton (cyclic)	:	Trim as desired for attitude reference changes during hover and climb out.
RPM switch	:	Set 100%.

NOTE

RPM shall be set to 102% during take-off and landing in the airspeed range from 0 knot to 60 knots and in hovering conditions.

NR/N2 speed	:	100% stabilized, check.

ROLLING TAKE-OFF

Collective and cyclic	:	Apply collective and forward cyclic as necessary to obtain forward speed on the ground. Apply collective as necessary to become airborne. Accelerate to 60 Knots and to desired climb.
Tail rotor pitch	:	Apply as necessary to maintain direction.
Flight instruments	:	Check.
Engine instruments	:	Within limits.
Main hydraulic system instruments	:	Within limits.
Utility hydraulic system instruments	:	Within limits.
Landing gear	:	UP (by 200 ft AGL). Check light sequence: - Green light off - Red light on (retraction) - Red light off (retracted and locked)

FOR TRAINING ONLY

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

MAIN UTIL CHRG caution message	:	Displayed during landing gear operation. Out after the landing gear is locked.
Utility hydraulic systems	:	Pressure within limits.
FTR push-button (cyclic)	:	Trim as desired for attitude reference changes during climb out.
RPM switch	:	Set 100%.

NOTE

RPM shall be set to 102% during take-off and landing in the airspeed range from 0 knot to 60 knots and in hovering condition.

NR/N2 speed	:	100% stabilized, check.

IN FLIGHT

NOTE

During IFR flight, the pilot is recommended to cross check the flight instruments indication for consistency, including the ADI stand-by indicator.

Collective	:	Adjust as necessary to keep engine parameters within limits.
Load Share (LD-SH) switch	:	As required.

NOTE

The LD-SH switch allows the pilot to keep matched engine torque or TOT, as required.

Airspeed	:	Maintain within limits shown on airspeed placard.
Landing light	:	OFF, if used. LANDING LT ON advisory message out.

FOR TRAINING ONLY

Pitot heat : As required.

CAUTION

Turn Pitot heat on for flight in visible moisture and in rain regardless of ambient temperature.

Altitude : As desired.

NOTE

Refer to applicable operating rules for high altitude oxygen requirements.

Landing gear lever : DOWN when flying below 200 ft height AGL.

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

HYD UTIL CHRG caution message : Displayed during landing gear operation.
Out after the landing gear is locked.

Utility hydraulic systems : Pressure within limits.

HELIPILOT must be monitored and re-centered by depressing FTR button.

FLIGHT DIRECTOR OPERATION (IF INSTALLED)

Attitude mode COUPLED/DECOUPLED (if installed)

The Flight Director may be used either coupled or uncoupled.
A COUPLED/DECOUPLED switch is provided for that purpose.

FOR TRAINING ONLY

When flying decoupled, the pilot flies the helicopter to satisfy Flight Director Commands as programmed on the Flight Director Control Panel.
When in this mode, the pilot should either center the commands or go to standby on his mode controller.
When flying coupled, the aircraft is automatically flown in pitch and roll to satisfy Flight Director Commands.

VOR operation

NAV MODE

CAUTION

VOR approaches in NAV mode are prohibited.
VOR approaches are permitted in VOR APP mode.

The NAV mode is designed for cross-country operation and localizer/glide slope approaches and any radial capture in this mode should be effectuated more than 15 NM from the station.
When flying near a VOR station the flight director senses the ″Cone of Confusion″ and automatically switches to Over Station Sensor Logic at which time course deviation signal is ignored and the system will fly course arrow heading for a fixed time to allow for station passage. This system can cause significant course deviation prior to termination of Over Station Sensor Logic; therefore it is recommended to use the HDG Mode near the station.

VOR APP MODE

The VOR APP Mode is designed for operating radial captures less than 15 NM from the station and to permit VOR approaches.
Very near to the station oscillations could occur around the tracked radial.
If the oscillations are undesirable, use HDG mode.

FOR TRAINING ONLY

GO-AROUND MODE

CAUTION

When operating in GO-AROUND mode, use caution to avoid exceeding engine power limits when satisfying the collective command bar.

Missed approach - attitude hold mode

Apply power to approximately maximum continuous power.
For approach speeds below 80 KIAS no initial pitch-up attitude adjustment is required. Re-center HELIPILOT indicators with FTR button.

Approach angle vs airspeed and rate of descent (Sea level - ISA)

Kts IAS (CAS) / R/D Ft/min	60 (67)	70 (74)	80 (81)	100 (100)	120 (116)
500	4.2	3.8	3.5	2.8	2.4
750	6.3	5.7	5.2	4.2	3.7
1000	8.4	7.6	7.0	5.6	4.9

NOTE

The steepest demonstrated approach slope is 8.4 degrees.

APPROACH AND LANDING

RPM switch : Set 102%.

NOTE

RPM shall be set to 102% during take-off and landing in the airspeed range from 0 knot to 60 knots and in hovering conditions.

FOR TRAINING ONLY

NR/N2 speed	:	102% stabilized, check.
Landing gear lever	:	DOWN. Check light sequence: - Red light on (extension) - Red light off - Green lights on (extended and locked). LANDING GEAR caution message suppressed, if previously activated.

FOR TRAINING ONLY

FOR TRAINING ONLY

NOTE

When descending below 150 feet AGL vocal message ″ONE FIFTY FEET″ is active regardless landing gear status. This message is suppressed if AWG switch is set to REGRADE.

CAUTION

Check the lever to be correctly engaged in the DOWN position to avoid inadvertent retraction of the landing gear.

CAUTION

Do not operate landing gear at speeds above 120 knots.
Do not fly with landing gear extended at speeds above 120 knots.

MAIN UTIL CHRG caution message	:	Displayed during landing gear operation. Out after the landing gear is locked.
Utility hydraulic systems	:	Pressure within limits.
Nose wheel lock	:	ON (lever up).
Parking brake	:	OFF (ON if landing on a slope).
External light	:	As required.
Approach path	:	During the approach, reduce the airspeed gradually with the cyclic and apply collective as necessary so that the helicopter will arrive at a hover in ground effect in such a manner that, in the event of an engine failure along the approach, it is possible to make a safe landing.

FOR TRAINING ONLY

After reaching a hover, descend slowly to the surface.
After ground contact lower the collective lever to the bottom stop, unless taxiing is desired, in which case maintain to control the helicopter.

CAUTION

The helicopter is free of ground resonance.
However if, for some reason, ground resonance should occur, lift the helicopter free of the ground immediately. If unable to become airborne, lower collective and shut-down engines.

Pedal brakes	: As necessary
Parking brake	: ON (OFF if ground taxiing is desired).
Nose wheel lock	: OFF if ground taxiing is desired.

NOTE

A run-on landing is possible on smooth surfaces. In this case the forward speed must be below 40 Knots and ground contact should be made at minimum vertical speed. Once ground contact is made, collective can be reduced, maintaining enough to control the helicopter. Forward speed can then be reduced to a taxiing speed or the helicopter may be brought to stop.

SHUTDOWN

Collective lever	: Check fully down.
Cyclic stick	: Centered and trimmed.
Pedals	: Centered.

FOR TRAINING ONLY

NOTE

Do not apply collective in this phase and during subsequent rotor deceleration, particularly in windy conditions.
Avoid any cyclic movement except to prevent hitting blade stops below 85% rotor RPM.

ENG 1 and 2 MODE switches : Set to IDLE then to OFF.

NOTE

If necessary the engine may be shut-down directly from FLT.

Fuel pumps 1 and 2 : OFF. FUEL PUMP 1 and 2 caution messages displayed.

CAUTION

During shutdown check that the N1 speed decelerates freely. Note any abnormal noise or rapid rundown and take corrective action as required per Maintenance Manual.

WARNING

If there is any evidence of fire within the engine after shut-down, perform immediately a DRY MOTORING RUN as described in this Section.

ENG l and 2 FUEL switches : OFF. Fuel valve indicators horizontal.

CROSS FEED switch : CLOSED. Crossfeed indicator vertical.

Pitot heat : OFF.

FOR TRAINING ONLY

Cockpit lights	: OFF.
External lights	: OFF.
Landing light	: OFF, if used.
Miscellaneous switches	: OFF.
RAD-MSTR (Radio Master) switch	: OFF.
Power levers	: OFF (full aft).

NOTE

Move both engine power levers to OFF before turning off the electrical power in order to avoid fuel spillage on the ground.

BATTery and GENerators 1 and 2	: OFF.
INVerters 1 and 2	: OFF.

NOTE

(with EDU software version 007 and DAU software version 006 only). Wait until N1 is 0% before turning off the electrical power to permit DCU to memorize the LCF counts.

POST FLIGHT CHECK

Landing gear lever lock	: NORMAL and lockwired.

If conditions require, perform the following:

Pitot, intake and exhaust covers	: Installed.

NOTE

The pitot tube covers must be installed at least 5 minutes after pitot heat has been switched off. The engines exhaust ducts covers must be installed at least 30 minutes after engine shut down. Refer to Maintenance Manual for additional information.

FOR TRAINING ONLY

FLIGHT HANDLING CHARACTERISTICS

STEEP APPROACHES AND VERTICAL DESCENT MANOEUVRES

Low speed steep approaches (up to 20 kts) and vertical descent manoeuvres should be performed with a rate of descent not exceeding 900 ft/min.

FOR TRAINING ONLY

FOR TRAINING ONLY